NHANES 2021–2023: Predicting Weight Change Among Adults Attempting Weight Loss

Data Science Boot Camp — Project Proposal Summary

1. Project Overview and Research Question

This project uses NHANES 2021–2023 data to examine whether demographic, socioeconomic, behavioral, and health-related characteristics can predict percentage weight change among adults aged 20 and older who reported attempting to lose weight during the previous 12 months.

The project is motivated by the gap between weight-loss attempts and actual weight-loss outcomes. From a goal-striving and self-regulation perspective, we are interested in identifying characteristics that distinguish individuals who experience greater weight loss from those who experience little weight change or weight gain.

Primary research question: How accurately can we predict percentage weight change among adults attempting weight loss, and which characteristics contribute most to predictive performance?

Because NHANES is cross-sectional, the project focuses on retrospective prediction of reported weight change rather than forecasting future weight loss.

2. Data Collection and Feature Engineering

The analytic sample includes **3,493 adults** who reported attempting weight loss and had valid percentage weight-change information.

The outcome is calculated using self-reported current weight and weight one year earlier:

Percentage Weight Change = [(Current Weight − Previous Weight) / Previous Weight] × 100

Sixteen initial predictors were developed across several domains:

- **Demographic and socioeconomic:** Age, sex, education, marital status, race/ethnicity, income-to-poverty ratio, and employment status.
- **Behavioral:** Moderate and vigorous leisure-time physical activity, sitting time, sleep duration, and smoking status.
- **Health-related:** Estimated baseline BMI, medical condition count, hypertension history, and high cholesterol history.

Additional features were engineered and evaluated, including depressive symptoms, current dieting behavior, alcohol consumption, and weekly working hours.

Missing values and questionnaire skip patterns were examined during data cleaning. Numerical and categorical preprocessing are performed within machine-learning pipelines.

3. Modeling and Preliminary Results

The analytic sample was divided into **2,794 training participants (80%)** and **699 held-out test participants (20%)**.

Three regression models were evaluated using five-fold cross-validation within the training data.

Model	MAE	RMSE	R^2
Dummy Regressor	6.463	9.457	-0.001
Linear Regression	6.189	8.781	0.137
Random Forest	6.178	8.769	0.138

Both Linear Regression and Random Forest improved predictive performance relative to the dummy baseline. However, Random Forest's advantage over Linear Regression was minimal.

Feature Selection

Permutation importance identified estimated baseline BMI as the strongest predictor, followed by income-to-poverty ratio, age, vigorous physical activity, and smoking status.

Comparisons of progressively larger predictor subsets indicated that adding demographic and behavioral information improved prediction beyond baseline BMI alone.

However, adding depressive symptoms, current dieting, weight-loss dieting, alcohol consumption, or weekly working hours individually did not meaningfully improve cross-validated performance. The original 16-predictor set remains our provisional benchmark.

4. Model Evaluation and Robustness

The evaluation strategy uses MAE, RMSE, and R^2, with preprocessing fitted within cross-validation training folds to minimize data leakage.

Two initial robustness findings are particularly relevant:

Baseline BMI sensitivity: Removing baseline BMI reduced Random Forest R^2 from 0.138 to −0.005, suggesting substantial dependence on baseline body size. Because baseline BMI and percentage weight change share a previous-weight measurement, mathematical coupling remains an important consideration.

Extreme weight changes: Among training participants, 135 individuals (4.83%) reported absolute weight changes greater than 20%. These participants accounted for 39.59% of total squared prediction error, indicating substantially poorer prediction for extreme weight changes.

All participants are retained in the primary analysis.

5. Remaining Work and Methodological Considerations

The next phase will focus on assessing cross-validation stability, evaluating subgroup prediction errors, and determining whether further feature refinement or alternative models improve performance.

A major methodological consideration is the NHANES complex sampling design. Current results are unweighted and use participant-level random splitting. The appropriate incorporation of survey weights, strata, and clustering into model evaluation remains to be determined.

The final test set will remain untouched until model selection is complete.

Current status: Baseline models, initial feature selection, a written evaluation plan, and preliminary robustness checks are complete. Results and documentation have been committed to the project's GitHub repository.